

None

THE ARENA HUB – MASTER CONCEPT REFERENCE

Version: v1.1.0_PILOT | Bible: v1.0_Pilot_Ready | 2026-03-26

SECTION 1 – ARENA HUB v1.0 (PILOT READY)

WHAT IT IS

A sovereign, school-controlled digital infrastructure for Alternative Provision (AP) and SEMH education. All data lives inside the school's own Google Workspace Tenant. No third-party storage. No external databases.

ARCHITECTURE: Sovereign Hybrid v7.0.0 – LOCKED

- BUCKET A: Curriculum Library (global, read-only for schools)
- BUCKET B: Performance & Welfare Data (school-specific, portable)
- SYSTEM LEVEL: Immutable audit infrastructure

FIVE PILLARS (all code-complete at pilot)

1. PLAYER CARD (PC) – student performance portal
Daily register, FIFA-style engagement scores, qualification tracking, opportunity reports, student chronology.
Deployed as: ARENA_PC_SCRIPTS (Web App, v4 LIVE)
2. WELFARE STATION (WS) – safeguarding + welfare
Incident logging, body map, SLT response hub, DSL settings, 2FA login via Twilio, export.
Deployed as: ARENA_WS_SCRIPTS (Web App, v6 LIVE)

3. KIT ROOM (KR) – curriculum & partner management
Kit/resource management, Partner Bible, Cloud Run ingestion pipeline for AI-assisted evidence scoring.
Deployed as: KIT_ROOM_SCRIPTS (Web App, v6 LIVE)
4. ARENA MANIFEST – Jonathan's admin console (MYSELF only)
School deployment, licence key generation, tier management, unboxing wizard, upgrade/downgrade modal.
Deployed as: ARENA_MANIFEST (Web App, v2 LIVE)
5. CORE LIBRARY (JB_ARENA_CORE_LIBRARY) – shared engine
ENGINE_CORE, AI_GATEWAY, TIMETABLE_ENGINE,
TAXONOMY_ENGINE, DFE_GATEWAY.
All PC/WS/KR projects call this via alias "Arena".

AI ENGINE

- Auto-approve threshold: 85%
- Human verify floor: 50-84%
- Auto-reject ceiling: $\leq 49\%$
- Batch page size: 500 records
- Node AI ID: PRV_AI_ND_001
- Cloud Run endpoint: europe-west2 (GCP:
speedy-elf-490219-c6)

SOVEREIGNTY RULES

- All PII stays in Bucket B Sheets (school's Drive)
- Every write records: WHO / WHAT / WHEN / WHERE (source tag)
- Drive gating: app locates BUCKET_B_DRIVE_ID via user email
- Forensic audit trail: RULE_VERSION_ID on every config entry

UNBOXING FLOW (new in S9)

Jonathan seeds a PC Master Template copy per school →
School opens sheet → MANIFEST_FORM.html wizard fires →
PC_MASTER_UNBOXING.gs creates folder, ground-zero backup,
copies data template, registers HT/DSL, writes back to
ARENA_MANIFEST, sets UNBOXED=TRUE flag.

TIER STRUCTURE

T1_PC | T2_PC_WS | T2_PC_KR | T3_FULL

VERSION CONTROL

Every ADMIN_CONFIG entry carries a VERSION_ID (format: CFG_[YYYYMMDD]_[SEQ_3]). Historical data always renders against the rules active at time of capture.

PILOT STATUS (as of 2026-03-26)

- ✓ All mirrors written
- ✓ Bible v1.0_Pilot_Ready generated
- ✓ P1-P5 Jonathan paste actions COMPLETE
- ✓ Testing begins 2026-03-27 23:00
- △ developmentMode: TRUE – flip to FALSE before go-live
- △ Library versions unpinned – pin at go-live

SECTION 2 – ARENA HUB v2.0 (INCLUSION & PORTABILITY)

Status: PLANNING_SHADOW_BUILD

THEME: Bring in the excluded. Make data portable.

OBJECTIVES

1. Lite Portal – lightweight interface for school refusers (low-bandwidth, minimal UI, same sovereign backend)
2. Parent/Home Portal – parents submit evidence directly (attendance photos, home learning, welfare observations)
3. Read-Only Cloud Mirror (Shadow Build) – real-time read-only mirror of v1.0 Sheets to Cloud Run backend. First cloud-native phase. No PII leaves Bucket B.
4. Parent Data Portability – automated GDPR Article 20 export requests fulfilled without Jonathan intervention.

TECHNICAL ADDITIONS

Parent Push Protocol

Direct-to-LA evidence hopper – parents submit, evidence lands in school's sovereign vault, auto-tagged with source.

Shadow Sync

Real-time mirroring of v1.0 Sheets → Cloud Run backend.
READ-ONLY. No writes back to Sheets from cloud layer.
Priority first step of cloud-native build.

Identity Bridge

Gmail/OAuth gating for external parent accounts.
Parents advised to create Google account (accepted at pilot
as sufficient – no separate auth layer needed yet).

SOVEREIGNTY GUARDRAILS (v2 additions)

Rule 1: No PII in central Cloud SQL – stay in Bucket B Sheets
Rule 2: App must locate BUCKET_B_DRIVE_ID via user email
Rule 3: Every entry records source tag: App / Classroom / Manual

SOCKET RESERVATION

PARENT_HUB_ID must be reserved in ADMIN_CONFIG before any v2 build begins. Plug-and-Play – must not touch v1.0.

NEW ENDPOINTS

/v2/lite – Lite Portal routing via PC_SHELL
(Parent portal endpoint TBD at design stage)

SECTION 3 – ARENA HUB v3.0 (FEDERATED REGIONAL NETWORK)
Status: BLUE_SKY_ARCHITECTURE

THEME: From school tool to regional infrastructure.

OBJECTIVES

1. Regional LA Hubs – Federated Spoke model. Each LA runs a Master Hub. Schools are sovereign spokes registered under their LA.
2. Google Classroom Integration – mainstream school data flows into the Arena without manual entry.
3. Regional Kit Manager – automated curriculum seeding pushed from LA Hub to school Kit Rooms.
4. Unified LA Reporting Heatmaps – cross-school attendance, welfare, and performance signals aggregated at LA level.

TECHNICAL ADDITIONS

Google Classroom Bridge

Translates Classroom grades + timestamps into Arena FIFA Velocity and Engagement scores.

OAuth scopes: `classroom.courses.readonly`,
`classroom.coursework.me`

Sovereignty: Classroom data shadow-synced into local Sovereign Vault – never stored centrally.

New endpoint: `/v3/classroom_sync` via `PC_SHELL`

Federated Handshake

Mechanism: Registry-based HTTPS GET/POST knocks

Registry: `Subordinate_Registry.json` hosted on LA Master Hub

Scaling: New sites added by `Deployment_ID` whitelisting

No central broker – each spoke calls its LA registry only.

Stateless Taxonomy Engine

`TAXONOMY_ENGINE` refactored to accept external LA-seeded JSON. Enables regional curriculum variation without forking the core library.

Namespace: `JB_arena_core_library` alias maintained throughout.

FEDERATION SOVEREIGNTY MODEL

Each school: owns its Bucket B, sovereign and portable
Each LA Hub: holds a registry of Deployment_IDs only –
no PII, no student data
Cross-LA: not in scope for v3

BUILDER RULES (all versions)

- v1.0 Pilot Ready is the immutable baseline
- All v2/v3 modules are Plug-and-Play – never break v1.0
- Shadow Sync (read-only) is the first cloud-native move
- PII never leaves Bucket B Sheets
- Every entry: source-tagged (App / Classroom / Manual)
- Library alias: Arena (JB_ARENA_CORE_LIBRARY) – always

END OF MASTER CONCEPT REFERENCE

State file updated. All P1–P5 confirmed complete. Testing slot locked: **27/3 23:00**.